

RayGlides 3W/4W EV Kit Cooling Controller - ESP32 GPIO Mapping (Optimized for ESP32-S3-N16R8)

This document provides a reference mapping of the ESP32-S3-N16R8 pins to the hardware interfaces, verified against your PlatformIO source code definitions.

1. GPIO Pin Allocations

ESP32-S3 Pin	Function / Net	Signal Type	Active Level / Details
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GPIO 1	BATTERY_VOLTAGE_PIN	Analog Input	ADC1_CH0 - Pack voltage sensing (Divider ratio 8)
GPIO 2	BATTERY_CURRENT_PIN	Analog Input	ADC1_CH1 - Current sensor midpoint 1.65V (mV)
GPIO 4	BATTERY_TEMP_PIN	Analog Input	ADC1_CH3 - Connected to LM35 output (LM35)
GPIO 5	SOLAR_VOLTAGE_PIN	Analog Input	ADC1_CH4 - Solar voltage input (Divider ratio 8)
GPIO 6	SOLAR_CURRENT_PIN	Analog Input	ADC1_CH5 - Solar current input
GPIO 7	GRID_BUTTON_PIN	Digital Input	Pulled high, button connects to GND
GPIO 8	MPPT_PWM_PIN	PWM Output	LEDC Channel 0, 5 kHz frequency, 8-bit resolution
GPIO 9	CAN_TX_PIN	CAN TX	Connected to TJA1050 CAN Transceiver TXD
GPIO 10	CAN_RX_PIN	CAN RX	Connected to TJA1050 CAN Transceiver RXD
GPIO 11	RS485_TX_PIN	UART TX	Connected to UART2 TX
GPIO 12	RS485_RX_PIN	UART RX	Connected to UART2 RX
GPIO 13	RS485_DE_PIN	Digital Output	RS485 Transceiver direction pin (DE/RE)
GPIO 14	RELAY_PIN	Digital Output	Active High - Powers external charging relay
GPIO 15	CHARGE_LED_PIN	Digital Output	Active High - Charging status indicator
GPIO 16	FAULT_LED_PIN	Digital Output	Active High - Fault status indicator
GPIO 18	FAN_PWM_PIN	PWM Output	LEDC Channel 1, 5 kHz frequency, 8-bit resolution
GPIO 19	NATIVE_USB_D-	USB Data	Direct differential native USB D- programming interface
GPIO 20	NATIVE_USB_D+	USB Data	Direct differential native USB D+ programming interface

2. Strapping Pin Audit (Safety Check)

- * **GPIO 0, 3, 45, 46** are hardware strapping pins on the ESP32-S3. They are bypassed or skipped in this layout:
- * `GPIO 3` (JTAG strapping pin) is skipped. `BATTERY\_TEMP\_PIN` was assigned to `GPIO 4` instead.
- * `GPIO 0`, `GPIO 45`, `GPIO 46` are unused in our configuration, ensuring clean bootloading.
- * **Octal PSRAM Constraints**: The ESP32-S3-N16R8 module uses Octal SPI PSRAM. All pins from `GPIO 26` through `GPIO 37` are reserved internally and must **never** be broken out or connected to external circuits. Our pinout strictly limits connections to the safe range `GPIO 1-20`, preventing memory collision crashes.

- * ****Native USB OTG Support****: Utilizing `GPIO 19` and `GPIO 20` enables direct USB flashing and JTAG debugging without needing external bridge ICs (like CP2102/CH340), reducing cost and PCB surface area.